



VOÏLA PLATFORM SPECIFICATION

Comprehensive Technical Reference: Features, KPIs, NLP/ML Algorithms, Agentic AI Architecture & Relational Schemas

Document Version:	v2.4.0 (Production Release)
Active Scale:	105,000 Ingested Interactions
Primary Tech Stack:	FastAPI, React + Vite, PostgreSQL, Qdrant, Snowflake, AWS Bedrock
Operational Domains:	Voice of Customer (VoC), SLA Triage, Root-Cause NLP, Agentic Copilot
Classification:	Engineering, Product & Governance Reference
Generated Date:	August 2026

Executive Summary:

Voilà is an enterprise-grade Voice-of-Customer Signal Intelligence and Agentic AI platform. It continuously ingests multi-million row customer support dialogue streams, maps conversational friction through NLP clustering, enforces operational SLA policies, and empowers support and product executives through an autonomous, grounded AI copilot.

1. Complete Platform Feature Catalog

1.1 Executive Dashboard & Operational Health (/)

- **Live Telemetry Grid:** Real-time tracking of 105,000+ interactions, 53.7% Resolution Rate, 133.7m Mean Response Time, 44.5% Reopen Rate, 24.2% Negative Friction Share, and 3.8/5.0 CSAT Index.
- **SLA Latency Distribution Matrix:** Dynamic bucketing of support response times (<15m, 15-60m, 1-4h, >4h) to immediately expose queue bottlenecks.
- **Sentiment & Friction Donut:** Polarity distribution calculating high-distress vs positive resolution sentiment.
- **Service Velocity Trend Analysis:** Temporal response curve visualizing queue backlog speed across hourly and daily windows.
- **Topic Quadrant Matrix (2D Scatter):** Multi-dimensional visualization plotting *Friction Severity* vs *Ticket Volume* to prioritize critical blockers over high-volume noise.
- **Unified Regional Intelligence:** Geographic benchmarking comparing North America, Europe, Asia Pacific, Latin America, and Middle East & Africa.
- **Priority Action Board:** Auto-generated P0, P1, and P2 action plans mapped directly to Engineering, Support Operations, and Billing teams.
- **Spike Detection Banner:** Real-time anomaly detection alerting teams when volume or negative tone spikes beyond normal thresholds.
- **Multi-Format Custom Export:** One-click executive report downloads in **PDF, CSV, JSON, and Markdown** formats.

1.2 Topic Clusters & Deep-Dive NLP (/topics)

- **Automated Topic Extraction:** BERTopic and TF-IDF clustering categorizing customer issues into semantic buckets.
- **Pre-Configured Complaint Clusters:** In-depth telemetry on *Poor customer support, Application malfunction, Delivery & Order tracking, Billing & Payment disputes, Network/Connectivity failures, and Account Access/2FA*.
- **Verbatim Customer Quotes:** Live anonymized dialogue evidence displaying raw customer complaints with timestamps and sentiment polarity.
- **Pain Score & Escalation Risk:** Weighted index quantifying operational risk based on repeat interactions and user sentiment.

1.3 Ask Data — Gemini / ChatGPT Conversational Copilot (/ask)

- **Modern Conversational Stream:** Clean Markdown rendering, Voilà gradient sparkle avatar, user bubble styling, animated reasoning dots, and quick prompt chips.
- **Persona & Dialogue Intelligence:** Natural answers for identity queries ('what is your name'), capabilities ('what can you do'), politeness ('thank you'), greetings, and operational policy guidance.
- **Root-Cause Volume Diagnosis:** Deep-dive explanations for specific volume queries (e.g. diagnosing the 15,700 messages driving support friction).
- **Action Toolbar & Telemetry Accordion:** Copy-to-clipboard button, feedback buttons, and collapsible visual KPI charts.
- **Query Audit History:** Persistent history sidebar with single-item deletion on hover and a global 'Clear All' action.

1.4 Streaming Ingestion & S3 Data Pipeline (/upload)

- **Multi-Million Row Support:** Handles huge CSV and Excel datasets with dual-mode streaming architecture.
- **In-Memory RAM Engine:** Sub-second streaming for small/medium dataset uploads with immediate KPI extraction.
- **Chunked Streaming Pipeline:** 20,000-row batch streaming with zero Out-Of-Memory (OOM) risk for massive files.
- **Dataset Catalog & Run Management:** Instant switching between dataset runs and a permanent delete confirmation modal.

1.5 Dataset Delta & Multi-Period Comparison (/compare)

- **Side-by-Side Variance:** Compare two uploaded runs or two calendar years (e.g. 2017 vs 2018).
- **Automated Delta Calculations:** Real-time calculations for Delta Resolution (%), Delta Response Time (mins), and Delta Friction.
- **Emergent Issue Detection:** Flags newly emerging complaint categories and tracks resolved friction clusters.

2. Operational Metrics & KPI Specifications

The platform calculates 8 primary and 6 derived operational indicators directly from normalized relational PostgreSQL tables:

Metric Name	Mathematical Formula	Baseline	Target	Operational Meaning
Resolution Rate (FCR)	$\frac{N_{\text{resolved}}}{N_{\text{total}}} \times 100\%$	53.7%	$\geq 75.0\%$	Percentage of interactions fully resolved on first inbound touchpoint.
Mean Response Time	$\frac{1}{N} \sum_{i=1}^N (T_{\text{reply}} - T_{\text{inbound}})$	133.7m	$\leq 15.0\text{m}$	Average time in minutes from customer inbound message to agent response.
Reopen Rate	$\frac{N_{\text{reopened}}}{N_{\text{resolved}}} \times 100\%$	44.5%	$\leq 20.0\%$	Percentage of tickets reopened by customer due to incomplete resolution.
Negative Friction Share	$\frac{N_{\text{sentiment} < -0.1}}{N_{\text{total}}} \times 100\%$	24.2%	$\leq 10.0\%$	Proportion of conversation volume expressing severe frustration or friction.
CSAT Index	$1.0 + 4.0 \times (0.6 \cdot \text{Res} + 0.4 \cdot (1 - \text{Neg}))$	3.8 / 5.0	≥ 4.5	Composite customer satisfaction index normalized to a 1.0 - 5.0 scale.
Topic Pain Score	$\text{Vol} \times \text{Neg} \times \left(1 + \frac{\text{Resp}}{100}\right)$	8,420 pts	$\leq 2,000$	Weighted severity scoring to prioritize engineering and support hotfixes.
Escalation Risk Index	$\frac{N_{\text{turns} \geq 3} + N_{\text{polarity} < -0.6}}{N_{\text{total}}} \times 100\%$	18.9%	$\leq 5.0\%$	Risk ratio of conversations escalating to supervisory intervention.

3. Machine Learning & NLP Algorithms

3.1 Semantic Topic Clustering (BERTopic & TF-IDF)

The platform leverages a hybrid NLP clustering pipeline:

- Text Normalization & Tokenization:** Strips metadata, usernames, URLs, and punctuation while preserving sentiment modifiers.
- Dense Embeddings:** Text chunks are mapped into high-dimensional vector representations using 384-dimensional sentence transformers (all-MiniLM-L6-v2).
- Class-based TF-IDF (c-TF-IDF):** Extracts domain keywords for each cluster by calculating $SW_{t, c} = \text{tf}_{t, c} \times \log\left(1 + \frac{A}{f_t}\right)$ where A is the total corpus volume.

3.2 Sentiment Polarity & Tone Classification

- Scoring Model:** Hybrid VADER + Transformer sentiment model outputting continuous polarity in the range $[-1.0, +1.0]$.
- Classification Thresholds:** Positive ($\geq +0.10$), Neutral ($-0.10 < s < +0.10$), Negative (≤ -0.10). Extreme distress tickets are flagged when polarity < -0.60 .

3.3 Anomaly & Spike Detection Algorithm

- **Z-Score Statistical Filtering:** Continuous rolling evaluation $Z = \frac{X_t - \mu_{24h}}{\sigma_{24h}}$. An operational spike alert is triggered whenever $Z > 2.50$ (99% confidence interval) indicating an unprecedented surge in ticket volume or negative sentiment.

4. Agentic AI Reasoning Architecture & Tools

Voià features a multi-tool autonomous agentic reasoning pipeline with zero ungrounded hallucinations:

Step	Component	Execution Responsibility
1	QueryValidator	Inspects input question for data sufficiency, dataset fields, and semantic intent.
2	DecisionEngine	Decides required toolchain (PostgreSQL, Qdrant Vector DB, Snowflake, Analytics Hub).
3	Tool Orchestrator	Executes parallel sub-millisecond database queries, full-text searches, and vector lookups.
4	ContextBuilder	Grounds telemetry facts, verbatim customer quotes, and SLA metrics into an executive prompt.
5	BedrockClient	Synthesizes executive reasoning via AWS Bedrock Mantle / Local deterministic fallback.
6	ResultValidator	Verifies data confidence (MEASURED / NO_DATA) and ensures zero metric hallucinations.

4.1 Dedicated Tool Ecosystem

- **PostgreSQL Fast Analytics Tool:** Executes sub-5ms indexed queries across 105k conversation records, computing dynamic aggregations, SLA distributions, and cluster stats.
- **Qdrant Vector DB Tool:** 384-dimensional cosine similarity semantic search with non-blocking socket checks (<0.15s) for instant semantic quote retrieval.
- **Snowflake Data Warehouse Tool:** Cloud warehouse integration with offline short-circuit safeguards preventing blocking timeouts.
- **AWS Bedrock LLM Client:** Grounded reasoning with 0.6s circuit-breaker and instant local executive synthesis.

4.2 Enforced Governance & Operational Policies

1. **First-Response SLA Velocity Policy:** Target \$!e 15.0\$ minutes. Dynamic deflection of high-negative-friction tickets directly to senior specialists.
2. **Reopen Mitigation Policy:** Mandatory 48-Hour Customer Confirmation Window before ticket closure to combat premature resolution.
3. **Severity P0/P1 Escalation Protocol:** Tickets with \$!ge 3\$ repeat interactions or sentiment \$< -0.6\$ trigger automated supervisor alerts.
4. **Grounded Data Integrity Policy:** Copilot operates strictly on verified database telemetry with zero invented numbers or hallucinations.

5. Relational Database Architecture (PostgreSQL)

The platform stores all raw dialogues, normalized metrics, and query logs in a high-performance relational schema:

Table Name	Primary Keys & Indexes	Purpose & Telemetry Stored
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dataset_runs	run_id (PK), uploaded_at	Master catalog of uploaded dataset runs, total row counts, and status.
customer_conversations	id (PK), run_id, timestamp	105,000+ raw and clean customer dialogues, author roles, and response times.
kpi_runs	id (PK), run_id, created_at	Pre-aggregated executive metrics (Resolution Rate, Response Time, CSAT).
kpi_topics	id (PK), run_id, cluster_name	NLP friction clusters, volume, pain scores, and escalation cases.
kpi_sentiment	id (PK), run_id, sentiment	Distribution counts and percentages for positive, neutral, and negative tones.
agent_conversations	id (PK), user_id, timestamp	Conversational Copilot query audit logs, query types, and AI answers.
agent_tools	id (PK), agent_conversation_id	Audit log mapping which specific tools were executed for every AI reply.